DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-500-02 — Ultrasonic thickness survey of 6"-P300-LS-2003

| | | | |
|---|---|---|---|
| **Area** | A500 Pipe rack | **Task** | UT Ultrasonic thickness survey |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | rack-bent:1200 — Pipe rack bent at E 100'-0" — E 100'-0"  N 0.000"  EL 122'-0" | | |
| **Equipment** | — | **Lines** | 6"-P300-LS-2003 |
| **Isometrics** | ISO-LS-2003 (owner's drawings, attached) | | |
| **Permits** | General work permit | | |

## Reason

Thickness survey of the steam header.

## Scope of work

1. Erect access and remove insulation at the TMLs where fitted (class S300, insulated).
2. Calibrate the thickness gauge on a step block covering 0.21875 in.
3. Take four readings 90° apart at each of the 23 TMLs (SW-WO-6).
4. Report readings against the owner's minimum wall; raise a spool replacement for any reading below it.

Affected items: 6"-P300-LS-2003/T:2"-P300-LS-3007, 6"-P300-LS-2003/p4, 6"-P300-LS-2003/T:1"-P300-HS-5003, 6"-P300-LS-2003/p6, 6"-P300-LS-2003/T:2"-P300-LS-3009, 6"-P300-LS-2003/p8, 6"-P300-LS-2003/T:1"-P300-HS-5008, 6"-P300-LS-2003/p10, 6"-P300-LS-2003/T:1"-P300-HS-5013, 6"-P300-LS-2003/p12, 6"-P300-LS-2003/T:1"-P300-HS-5018, 6"-P300-LS-2003/p14, 6"-P300-LS-2003/T:1"-P300-HS-5023, 6"-P300-LS-2003/p16, 6"-P300-LS-2003/p17, 6"-P300-LS-2003/T:1"-P300-HS-5028, 6"-P300-LS-2003/p19, 6"-P300-LS-2003/T:1"-P300-HS-5033, 6"-P300-LS-2003/p21, 6"-P300-LS-2003/T:1"-P300-HS-5038, 6"-P300-LS-2003/T:3/4"-P150-V-8009, 6"-P300-LS-2003/T:3/4"-P150-D-8008, 6"-P300-LS-2003/T:3/4"-P150-D-8007

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-WO-1, SW-WO-2, SW-WO-6, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-LS-2003 | 6"-P300-LS-2003 | 6"-P300-LS-2003 | 7 | 0 error(s), 2 warning(s) — see below |

ISO-LS-2003 (PDF): ../isometrics-imported/ISO-LS-2003.pdf — the sheets that hold the work follow the ITP.

| Drawing | Sheet | Level | Finding |
|---|---|---|---|
| ISO-LS-2003 | 6 | warning | branch weld 3/4"-P150-V-8009/W1 (E 175'-6" N -1'-0" EL 123'-5", sheet 6) has no label |
| ISO-LS-2003 | 7 | warning | branch weld 3/4"-P150-D-8007/W1 (E 178'-0" N -1'-0" EL 123'-5", sheet 7) has no label |

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| rack-bent:1200 | structure | Pipe rack bent at E 100'-0" | | E 100'-0"  N 0.000"  EL 122'-0" |
| 6"-P300-LS-2003/T:2"-P300-LS-3007 | item | TEE REDUCING 6" × 2" P300 | 6"-P300-LS-2003 | E 22'-0"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p4 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 23'-9"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5003 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 25'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p6 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 30'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:2"-P300-LS-3009 | item | TEE REDUCING 6" × 2" P300 | 6"-P300-LS-2003 | E 35'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p8 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 40'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5008 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 45'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p10 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 55'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5013 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 65'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p12 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 75'-0"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5018 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 84'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p14 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 95'-0"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5023 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 105'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p16 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 115'-8-1/2"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p17 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 135'-3-1/2"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5028 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 145'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/p19 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 155'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5033 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 165'-6"  N -12"  EL 123'-0" |

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| 6"-P300-LS-2003/p21 | item | PIPE 6" P300 BEVELLED ENDS | 6"-P300-LS-2003 | E 170'-0"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:1"-P300-HS-5038 | item | TEE REDUCING 6" × 1" P300 | 6"-P300-LS-2003 | E 174'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:3/4"-P150-V-8009 | item | TEE REDUCING 6" × 3/4" P300 | 6"-P300-LS-2003 | E 175'-6"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:3/4"-P150-D-8008 | item | TEE REDUCING 6" × 3/4" P300 | 6"-P300-LS-2003 | E 177'-0"  N -12"  EL 123'-0" |
| 6"-P300-LS-2003/T:3/4"-P150-D-8007 | item | TEE REDUCING 6" × 3/4" P300 | 6"-P300-LS-2003 | E 178'-0"  N -12"  EL 123'-0" |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-500-02-I1 | Ultrasonic thickness at 23 TMLs, four readings each | Nominal wall 0.21875 in; each reading above the owner's minimum wall | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-500-02-TML01 | 6"-P300-LS-2003/T:2"-P300-LS-3007 | Tee 6"-P300-LS-2003/T:2"-P300-LS-3007 branch outlet | E 22'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML02 | 6"-P300-LS-2003/p4 | Pipe 6"-P300-LS-2003/p4 mid-length | E 23'-9"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML03 | 6"-P300-LS-2003/T:1"-P300-HS-5003 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5003 branch outlet | E 25'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML04 | 6"-P300-LS-2003/p6 | Pipe 6"-P300-LS-2003/p6 mid-length | E 30'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML05 | 6"-P300-LS-2003/T:2"-P300-LS-3009 | Tee 6"-P300-LS-2003/T:2"-P300-LS-3009 branch outlet | E 35'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML06 | 6"-P300-LS-2003/p8 | Pipe 6"-P300-LS-2003/p8 mid-length | E 40'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML07 | 6"-P300-LS-2003/T:1"-P300-HS-5008 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5008 branch outlet | E 45'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML08 | 6"-P300-LS-2003/p10 | Pipe 6"-P300-LS-2003/p10 mid-length | E 55'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML09 | 6"-P300-LS-2003/T:1"-P300-HS-5013 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5013 branch outlet | E 65'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML10 | 6"-P300-LS-2003/p12 | Pipe 6"-P300-LS-2003/p12 mid-length | E 75'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML11 | 6"-P300-LS-2003/T:1"-P300-HS-5018 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5018 branch outlet | E 84'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML12 | 6"-P300-LS-2003/p14 | Pipe 6"-P300-LS-2003/p14 mid-length | E 95'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML13 | 6"-P300-LS-2003/T:1"-P300-HS-5023 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5023 branch outlet | E 105'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML14 | 6"-P300-LS-2003/p16 | Pipe 6"-P300-LS-2003/p16 mid-length | E 115'-8-1/2"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML15 | 6"-P300-LS-2003/p17 | Pipe 6"-P300-LS-2003/p17 mid-length | E 135'-3-1/2"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML16 | 6"-P300-LS-2003/T:1"-P300-HS-5028 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5028 branch outlet | E 145'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML17 | 6"-P300-LS-2003/p19 | Pipe 6"-P300-LS-2003/p19 mid-length | E 155'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML18 | 6"-P300-LS-2003/T:1"-P300-HS-5033 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5033 branch outlet | E 165'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML19 | 6"-P300-LS-2003/p21 | Pipe 6"-P300-LS-2003/p21 mid-length | E 170'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML20 | 6"-P300-LS-2003/T:1"-P300-HS-5038 | Tee 6"-P300-LS-2003/T:1"-P300-HS-5038 branch outlet | E 174'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML21 | 6"-P300-LS-2003/T:3/4"-P150-V-8009 | Tee 6"-P300-LS-2003/T:3/4"-P150-V-8009 branch outlet | E 175'-6"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML22 | 6"-P300-LS-2003/T:3/4"-P150-D-8008 | Tee 6"-P300-LS-2003/T:3/4"-P150-D-8008 branch outlet | E 177'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-500-02-TML23 | 6"-P300-LS-2003/T:3/4"-P150-D-8007 | Tee 6"-P300-LS-2003/T:3/4"-P150-D-8007 branch outlet | E 178'-0"  N -12"  EL 123'-0" | 0.21875 in | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Inspection and test plan (SW-ITP-1..3)

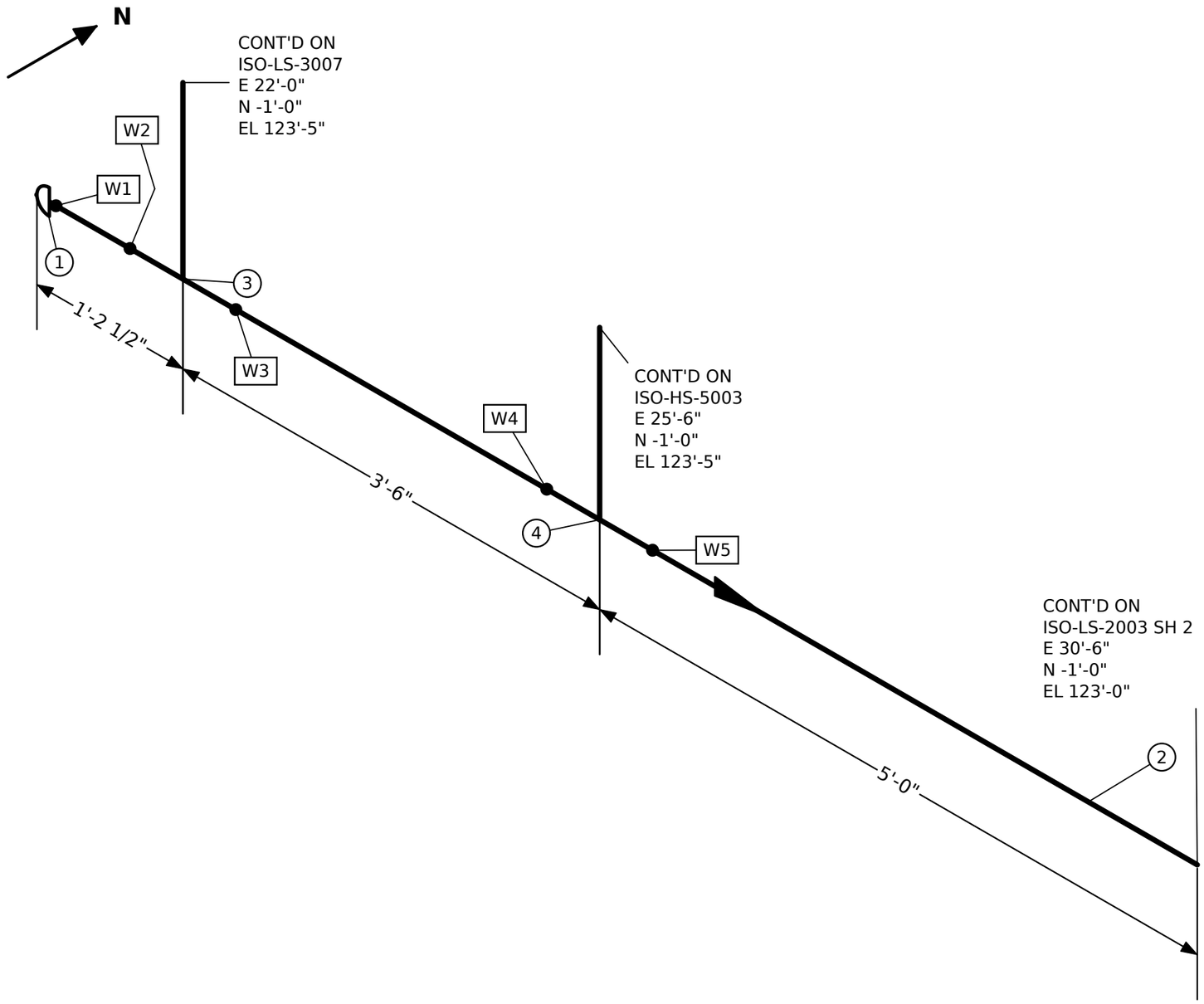
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Gauge calibrated on a step block; operator qualification checked | Owner UT procedure | Calibration within the owner's tolerance | Calibration record | R | R | R | | | | |
| 2 | Readings at 23 TMLs | SW-WO-6 | Four readings per TML recorded | Thickness record per TML | S | S | S | | | | |
| 3 | Results reviewed against the owner's minimum wall | Owner integrity procedure | Every reading above the minimum, or a work order raised | Signed thickness report | R | H | R | | | | |
| 4 | Insulation and access reinstated; work order handed back | SW-ITP-3 | All activities accepted | Signed ITP | H | H | R | | | | |

## Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

| Tag | Role | Drawing | Sheet | Weld label |
|---|---|---|---|---|
| 6"-P300-LS-2003/T:2"-P300-LS-3007 | affected | ISO-LS-2003 | 1 | |
| 6"-P300-LS-2003/p4 | affected | ISO-LS-2003 | 1 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5003 | affected | ISO-LS-2003 | 1 | |
| 6"-P300-LS-2003/p6 | affected | ISO-LS-2003 | 1 | |
| 6"-P300-LS-2003/T:2"-P300-LS-3009 | affected | ISO-LS-2003 | 2 | |
| 6"-P300-LS-2003/p8 | affected | ISO-LS-2003 | 2 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5008 | affected | ISO-LS-2003 | 2 | |
| 6"-P300-LS-2003/p10 | affected | ISO-LS-2003 | 3 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5013 | affected | ISO-LS-2003 | 3 | |
| 6"-P300-LS-2003/p12 | affected | ISO-LS-2003 | 3 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5018 | affected | ISO-LS-2003 | 3 | |
| 6"-P300-LS-2003/p14 | affected | ISO-LS-2003 | 4 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5023 | affected | ISO-LS-2003 | 4 | |
| 6"-P300-LS-2003/p16 | affected | ISO-LS-2003 | 4 | |
| 6"-P300-LS-2003/p17 | affected | ISO-LS-2003 | 5 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5028 | affected | ISO-LS-2003 | 5 | |
| 6"-P300-LS-2003/p19 | affected | ISO-LS-2003 | 5 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5033 | affected | ISO-LS-2003 | 5 | |
| 6"-P300-LS-2003/p21 | affected | ISO-LS-2003 | 5 | |
| 6"-P300-LS-2003/T:1"-P300-HS-5038 | affected | ISO-LS-2003 | 6 | |
| 6"-P300-LS-2003/T:3/4"-P150-V-8009 | affected | ISO-LS-2003 | 6 | |
| 6"-P300-LS-2003/T:3/4"-P150-D-8008 | affected | ISO-LS-2003 | 7 | |
| 6"-P300-LS-2003/T:3/4"-P150-D-8007 | affected | ISO-LS-2003 | 7 | |




**BILL OF MATERIALS**

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 1 | 6" | CAP 6" P300 |
| 2 | 7'-10" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 3 | 1 | 6"x2" | TEE REDUCING 6" × 2" P300 |
| 4 | 1 | 6"x1" | TEE REDUCING 6" × 1" P300 |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

COMPANY

**SMARTWELD**

TITLE

**LOW-PRESSURE STEAM**

**NOTES:**

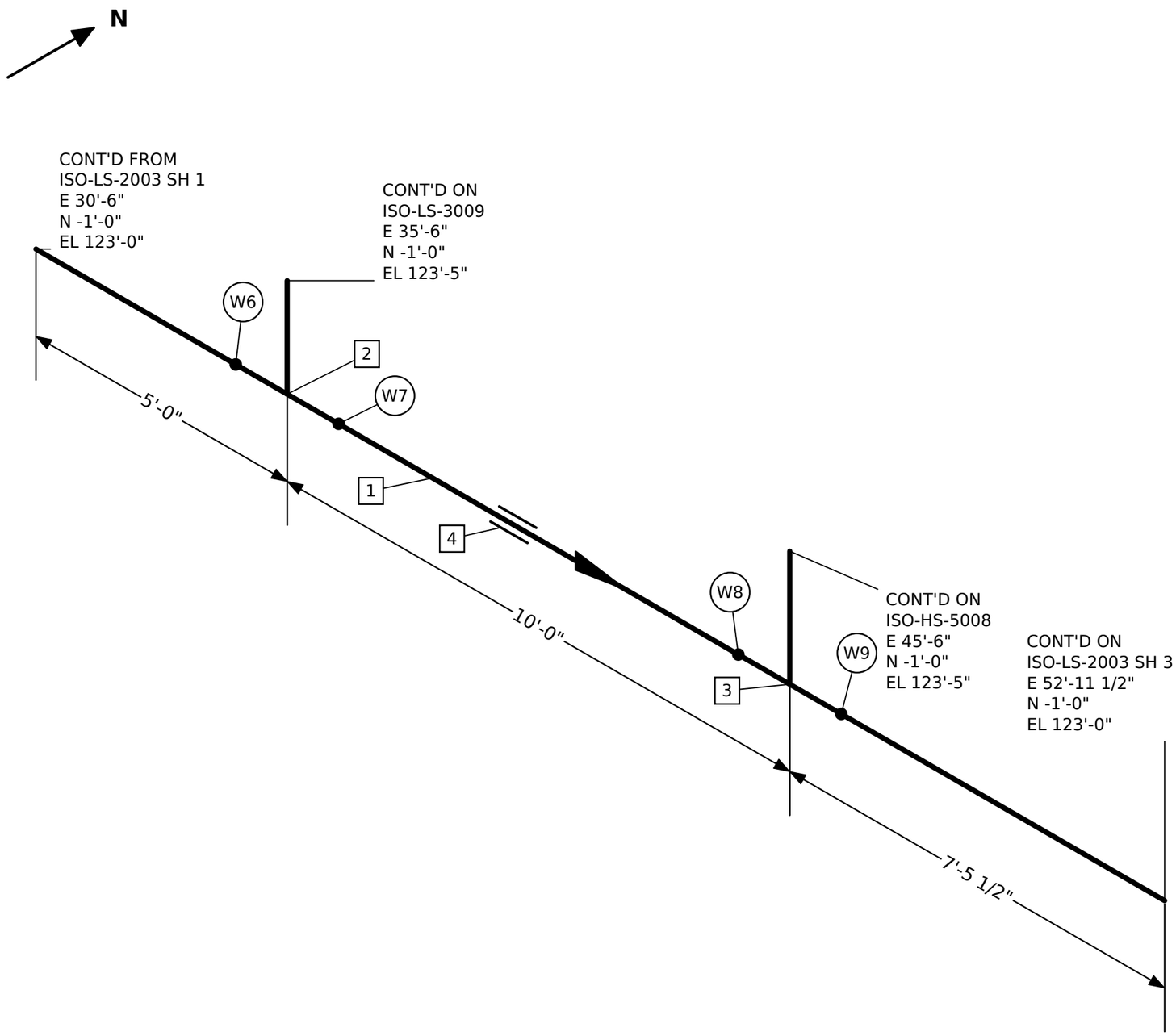
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 1 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




**BILL OF MATERIALS**

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 20'-9 1/2" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 2 | 1 | 6"x2" | TEE REDUCING 6" × 2" P300 |
| 3 | 1 | 6"x1" | TEE REDUCING 6" × 1" P300 |
| 4 | 1 | 6" | GUIDE FOR 6" LINE (GUIDE-6) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

**NOTES:**
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

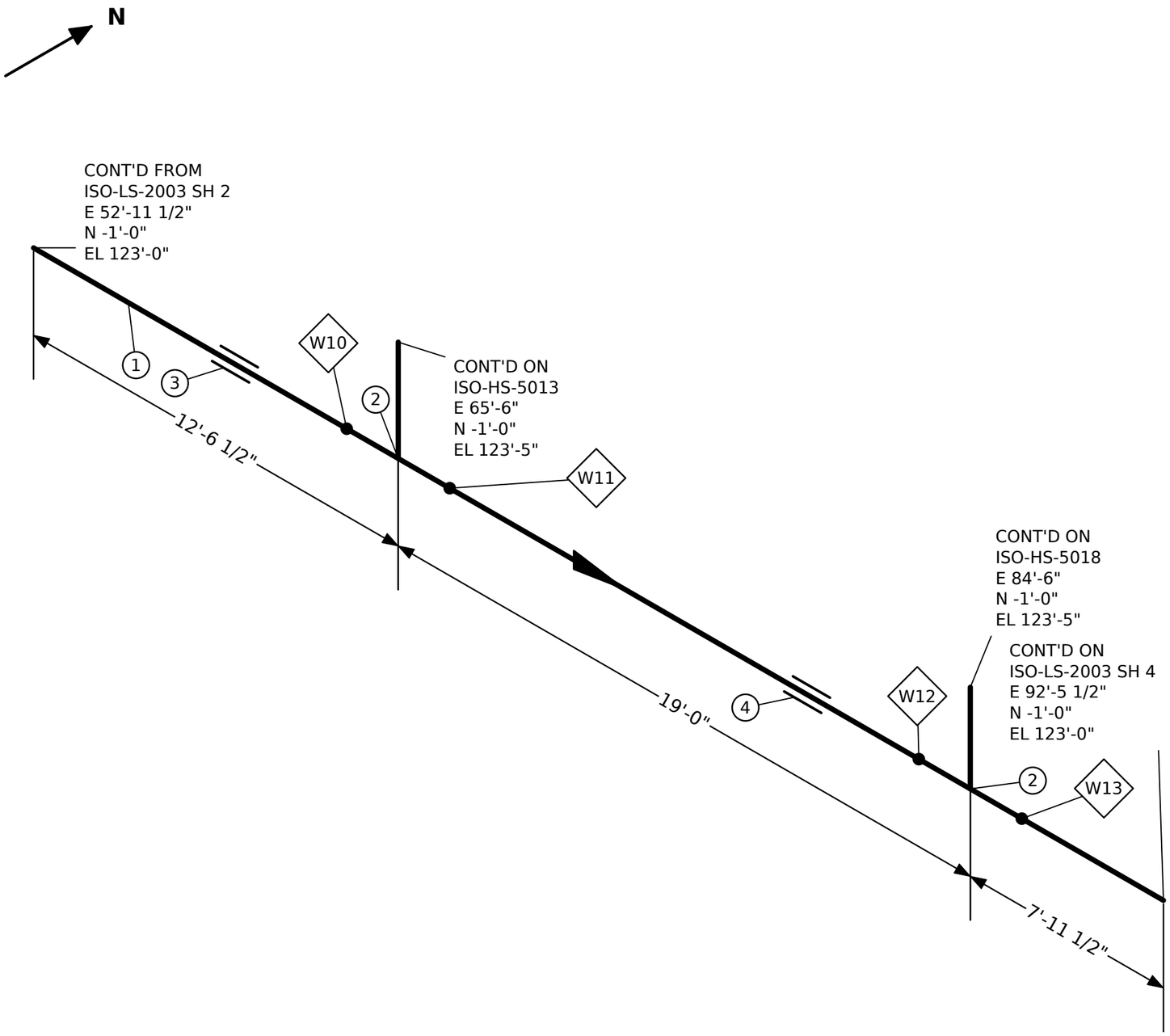
COMPANY
**SMARTWELD**

TITLE
**LOW-PRESSURE STEAM**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 2 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




**BILL OF MATERIALS**

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 37'-10" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 2 | 2 | 6"x1" | TEE REDUCING 6" × 1" P300 |
| 3 | 1 | 6" | SHOE FOR 6" LINE (SHOE-6) |
| 4 | 1 | 6" | GUIDE FOR 6" LINE (GUIDE-6) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. CHECK ORIENTATION BEFORE WELDING

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

COMPANY

**SMARTWELD**

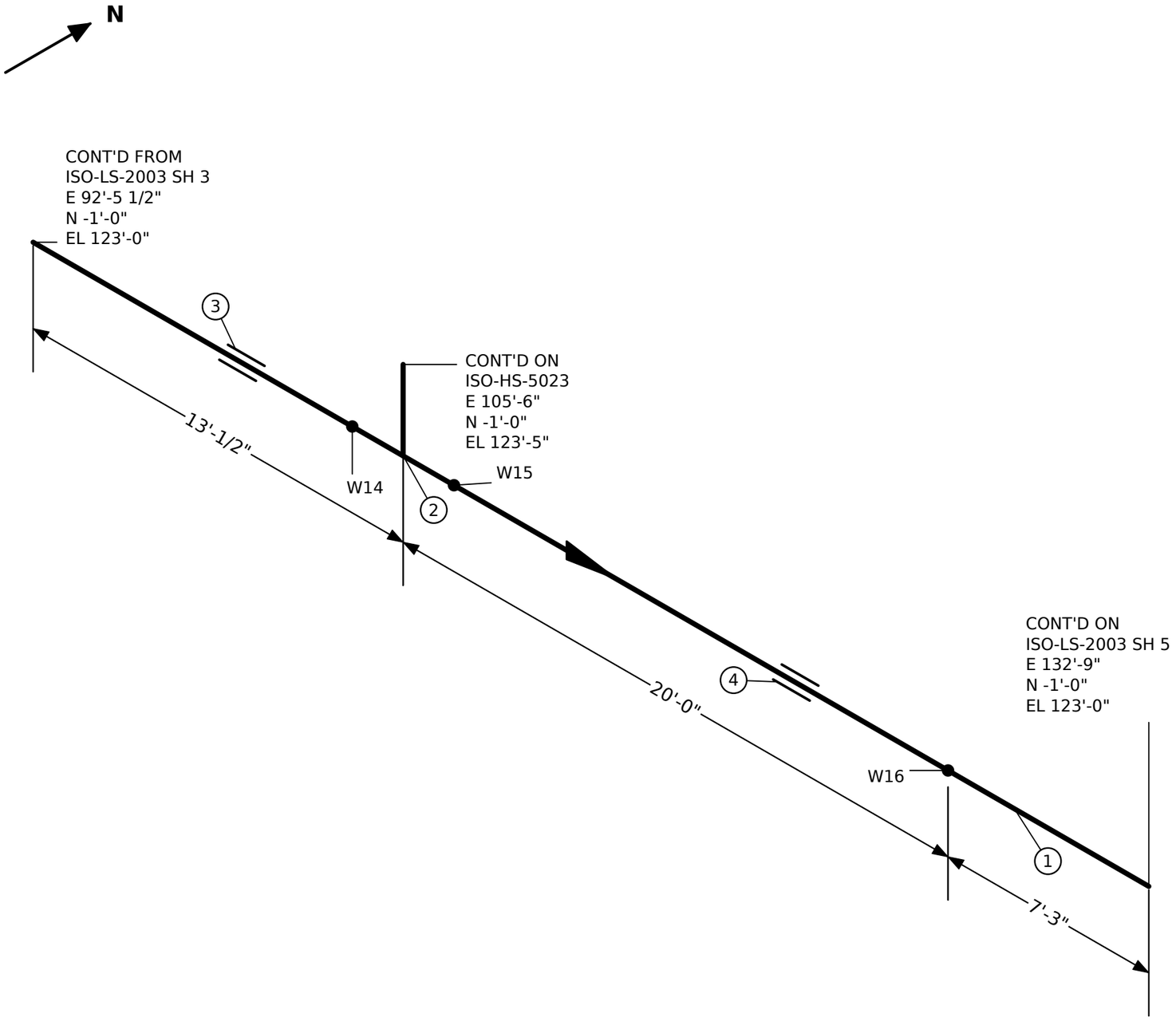
TITLE

**LOW-PRESSURE STEAM**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 3 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




**BILL OF MATERIALS**

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 39'-5 1/2" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 2 | 1 | 6"x1" | TEE REDUCING 6" × 1" P300 |
| 3 | 1 | 6" | ANCHOR FOR 6" LINE (ANCHOR-6) |
| 4 | 1 | 6" | GUIDE FOR 6" LINE (GUIDE-6) |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

COMPANY

**SMARTWELD**

TITLE

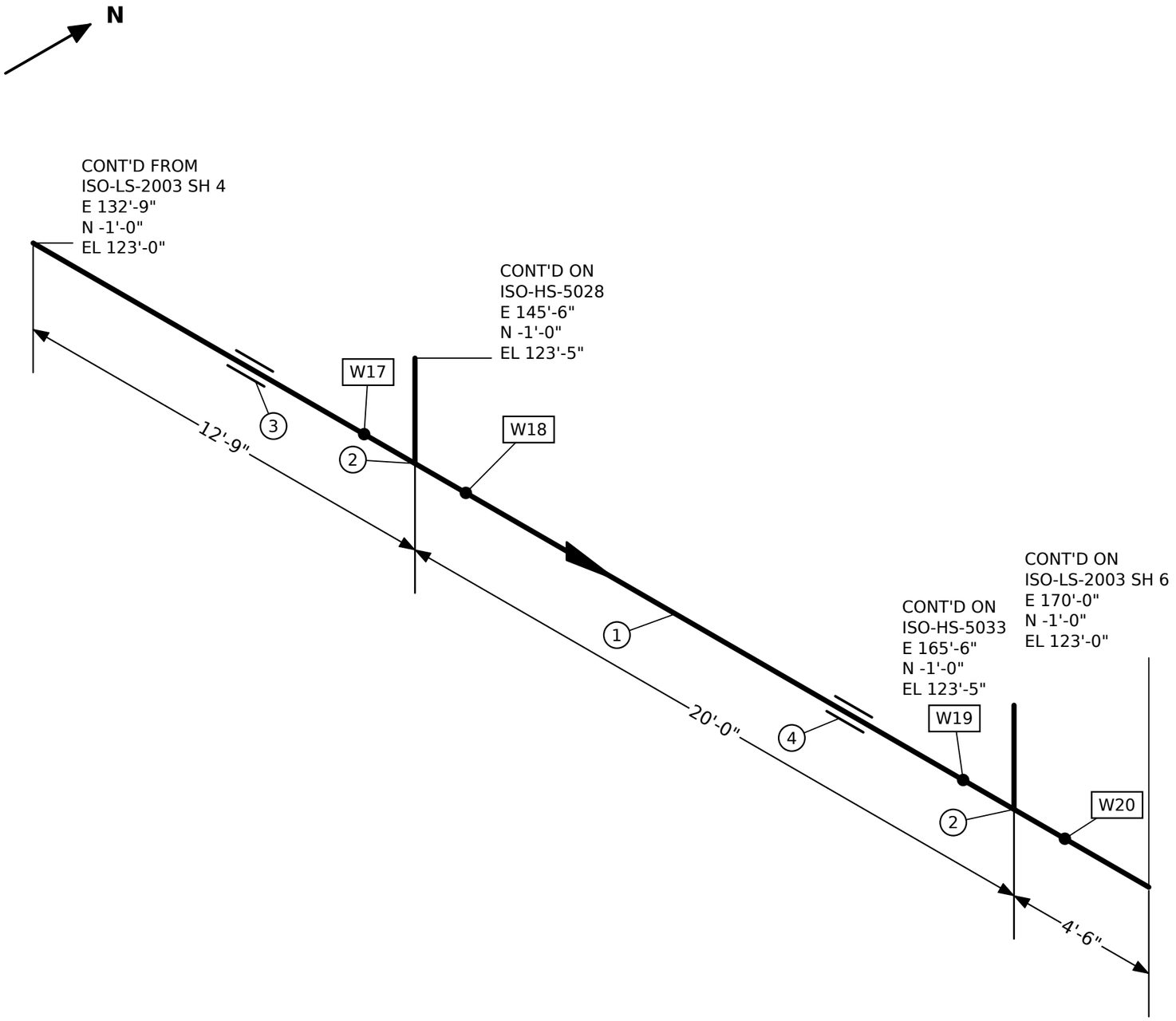
**LOW-PRESSURE STEAM**

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 4 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




**BILL OF MATERIALS**

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 35'-7" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 2 | 2 | 6"x1" | TEE REDUCING 6" × 1" P300 |
| 3 | 1 | 6" | SHOE FOR 6" LINE (SHOE-6) |
| 4 | 1 | 6" | GUIDE FOR 6" LINE (GUIDE-6) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

COMPANY

**SMARTWELD**

TITLE

**LOW-PRESSURE STEAM**

**NOTES:**

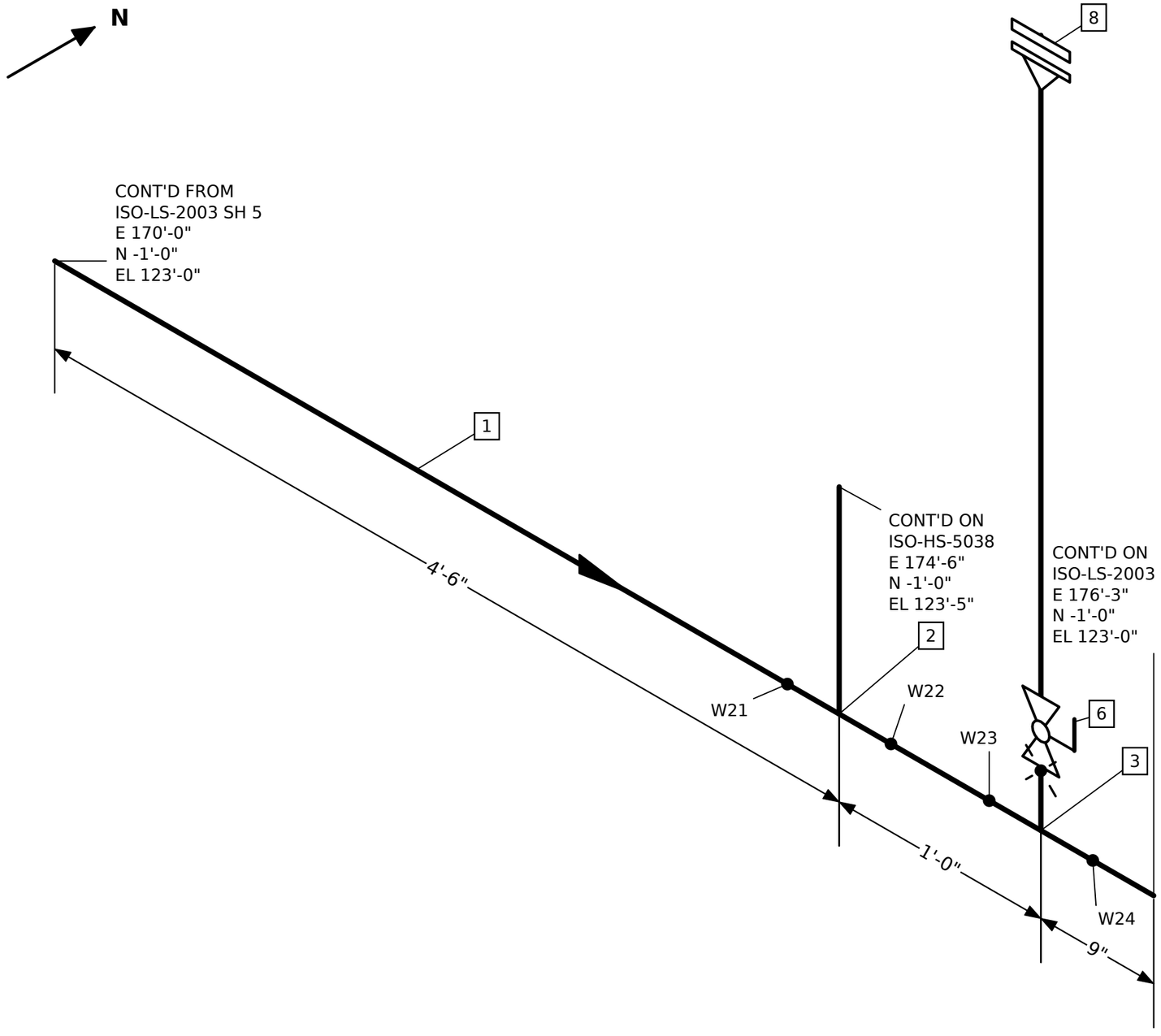
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 5 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 4'-7" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 2 | 1 | 6"x1" | TEE REDUCING 6" × 1" P300 |
| 3 | 1 | 6"x3/4" | TEE REDUCING 6" × 3/4" P300 |
| 4 | 1 | 3/4" | FLANGE WELD-NECK 3/4" P150 RAISED FACE |
| 5 | 2 | 3/4" | GASKET 3/4" P150 COMPRESSED SHEET |
| 6 | 1 | 3/4" | VALVE, BALL, FLOATING 3/4" P150 |
| 7 | 2 | 3/4" | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150 |
| 8 | 1 | 3/4" | FLANGE BLIND 3/4" P150 RAISED FACE |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

**NOTES:**
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

COMPANY

**SMARTWELD**

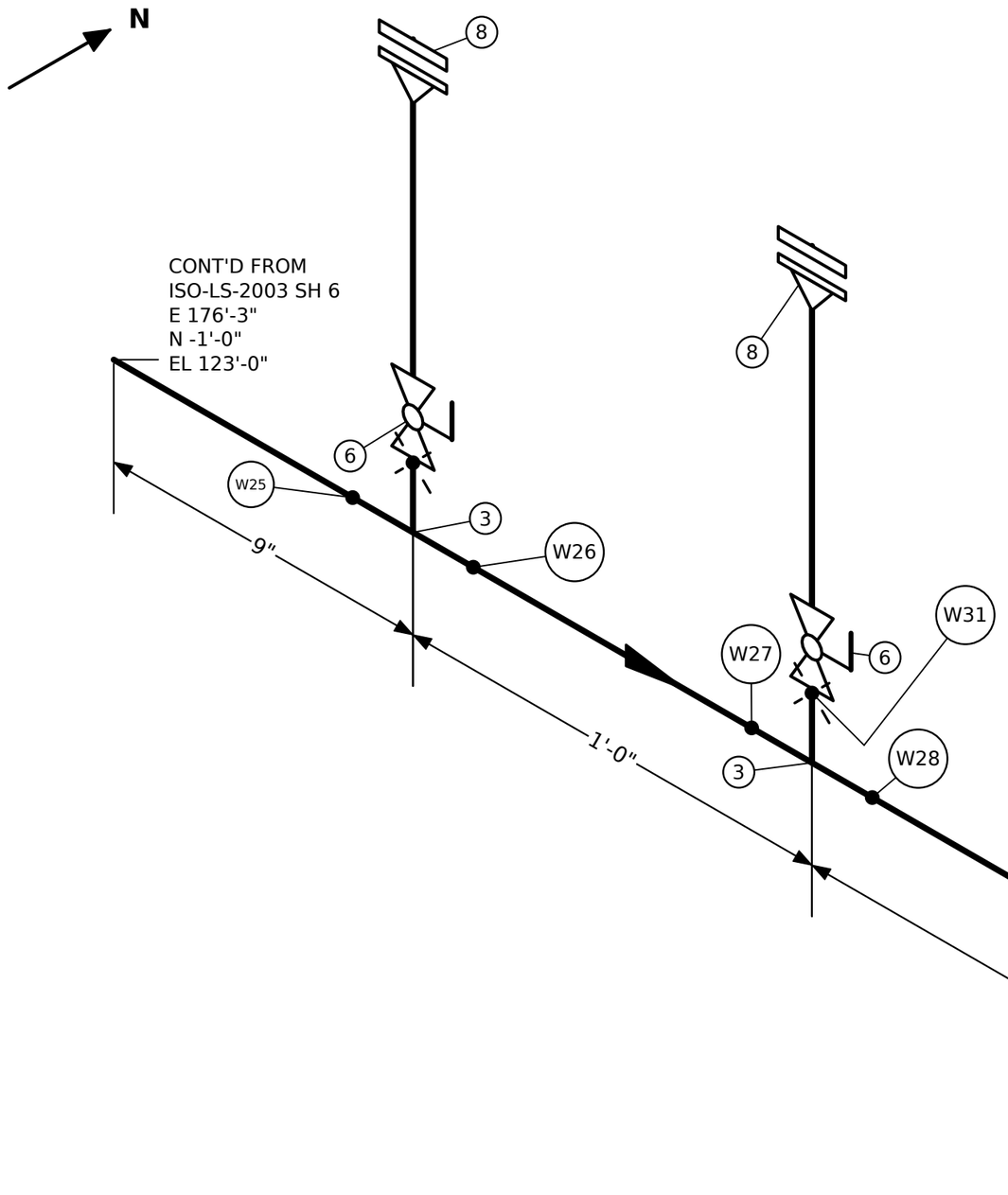
TITLE

**LOW-PRESSURE STEAM**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 6 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 1 | 6" | CAP 6" P300 |
| 2 | 1'-1" | 6" | PIPE 6" P300 BEVELLED ENDS |
| 3 | 2 | 6"x3/4" | TEE REDUCING 6" × 3/4" P300 |
| 4 | 2 | 3/4" | FLANGE WELD-NECK 3/4" P150 RAISED FACE |
| 5 | 4 | 3/4" | GASKET 3/4" P150 COMPRESSED SHEET |
| 6 | 2 | 3/4" | VALVE, BALL, FLOATING 3/4" P150 |
| 7 | 4 | 3/4" | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150 |
| 8 | 2 | 3/4" | FLANGE BLIND 3/4" P150 RAISED FACE |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |

COMPANY

**SMARTWELD**

TITLE

**LOW-PRESSURE STEAM**

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |

| PWHT | NDE | FLUID |
|---|---|---|
| - | - | LS |

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 6"-P300-LS-2003 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-2003 | 7 OF 7 | 0 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |

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